

Submission for the 2026 IAOS Prize for Young Statisticians

**Estimation of Job Vacancy Statistics Under Misclassification and Missing Data
Using Capture-Recapture**

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Abstract

Estimating the number of vacancies helps employers, businesses, and policymakers analyze labor demand. However, current methods usually rely on survey sampling or administrative registers, which often suffer from non-response and under-reporting. This paper presents a proof-of-concept approach for estimating the number of vacancies based solely on online data and administrative registers. Online job advertisements often lack information about occupation codes or desirable skills, so these attributes must be inferred. Therefore, the proposed method corrects for the bias associated with classification errors in categorical covariates. Additionally, considering that job offers may not include the exact number of advertised vacancies, the developed algorithm accommodates missing and censored data. The proposed approach is tested in a simulation study that utilizes actual data from the Polish public register-based job portal ePraca. While applied to data from Poland, the method can be easily adjusted for any other country and context. The simulation results show that the developed algorithm outperforms a naïve two-step approach by substantially reducing estimation bias.

1 Introduction

Understanding the size and characteristics of the demand for labor helps employers, businesses, and economic policymakers navigate the market. A fundamental part of this process is the precise estimation of the number of vacancies. A *job vacancy* is defined by Eurostat (2025) as a paid post that is newly created, unoccupied, or about to become vacant (note that this definition may differ between countries):

- for which the employer is taking active steps and is prepared to take further steps to find a suitable candidate from outside the enterprise concerned; and
- which the employer intends to fill either immediately or within a specific period of time.

Measuring labor demand requires complementing employment statistics with vacancy statistics (Davis et al., 2013). Vacancy data can also be used to study mismatch between job seekers and available jobs across occupations and sectors (Şahin et al., 2014).

Official methods for estimating the number of vacancies differ between countries. For details regarding the European Union, see Eurostat (2025). For example, in Czechia, the analysis is based solely on data from the administrative registers of Labor Offices. However, since January 1, 2012, it is no longer compulsory to report job vacancies to the Labor Office in Czechia. Although there has been no decline in reported job vacancies, this method might not provide a fully comprehensive picture of labor demand.

In Poland, the Statistical Office in Bydgoszcz provides quarterly reports on the number of vacancies in the country (as well as separately by territorial units and economic sectors) in the publication *The Demand for Labor* (DL; Statistics Poland, 2024). The study is based on survey sampling. This method has solid mathematical and theoretical foundations. However, Beręsewicz et al. (2021) compared survey data with administrative data and showed that DL underestimated the number of entities in Poland with at least one vacancy by about 10–15%. The reasons behind this phenomenon are increasing non-response and under-reporting rates, i.e., entities either refuse to answer the questions in the survey form or falsely report having no vacancies. For instance, in 2018, the response rates in DL ranged between 62% and 64% (Beręsewicz et al., 2021; Eurostat, 2025). Furthermore, the accompanying issue of under-reporting in official surveys was also shown in the Dutch Road Freight and Transport Survey (Klingwort, Buelens, et al., 2021; Klingwort, Burger, et al., 2021).

These examples highlight the need for alternative data sources to reliably estimate the number of vacancies. One can consider data collected from online job advertisements (OJAs; Kureková et al., 2015; Turrell et al., 2019). However, despite the large size of such datasets, OJAs represent only a part of the target population (Napierała et al., 2022). There is no single Internet portal that collects all job offers.

To overcome this issue, one can use capture-recapture (CRC) methods, which were originally developed to estimate the unknown size of an animal population (International Working Group for Disease Monitoring and Forecasting, 1995). They are also now applied in the social sciences and official statistics, allowing researchers to estimate the target quantity after integrating several data sources. For example, Klingwort et al. (2021) used CRC methods to estimate the total shipment weight transported by trucks in the Netherlands.

The CRC framework applied to job vacancy estimation is conceptually illustrated in Figure 1. The overlapping circles represent job offers identified across three distinct data sources. By modeling the overlapping areas, CRC methods estimate the number of job offers not captured by any job portal.

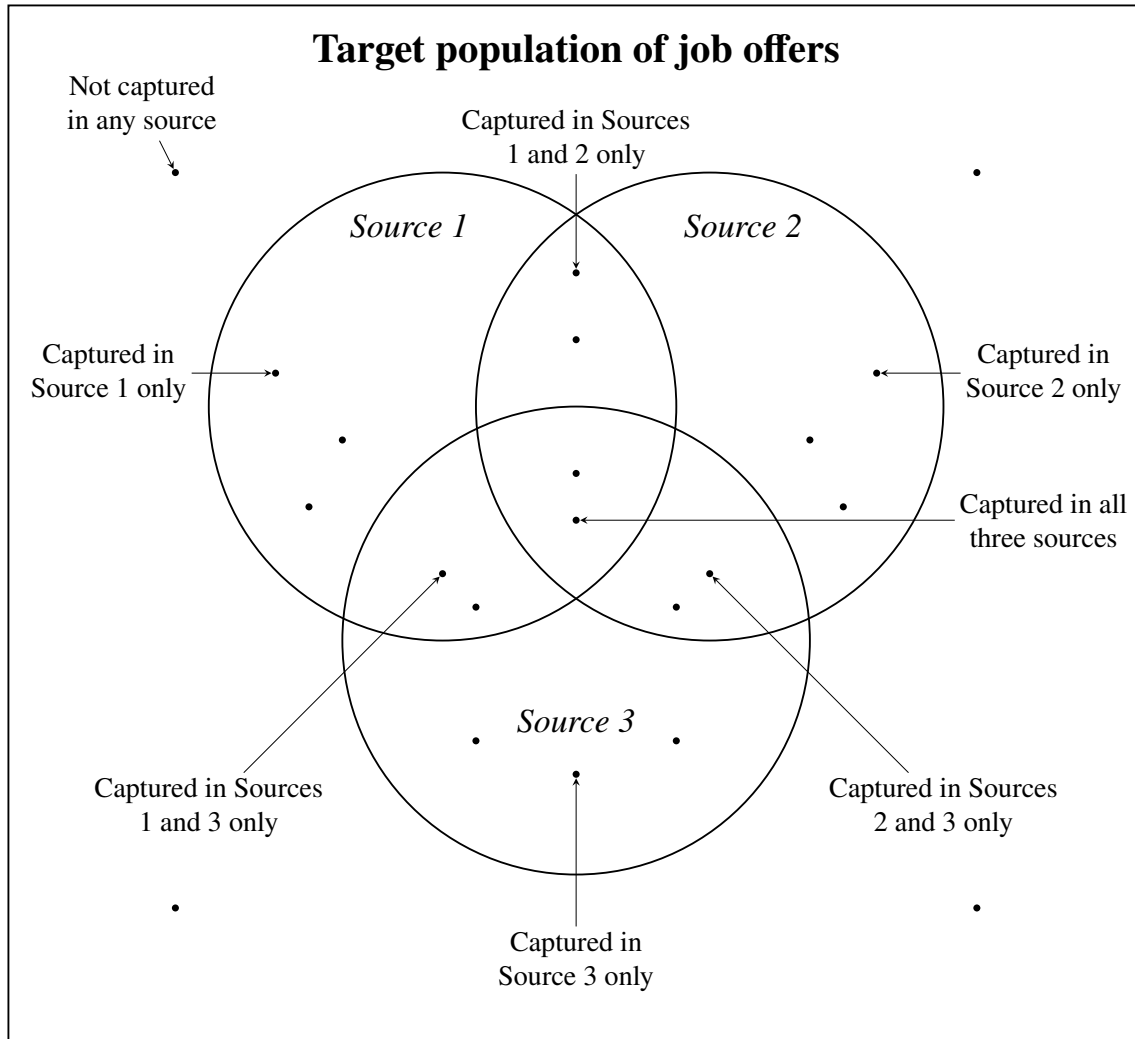


Figure 1: Conceptual illustration of the capture-recapture framework for three data sources applied to job vacancy estimation.

This paper presents a proof-of-concept approach for estimating the number of job vacancies based on OJAs and administrative registers. The method takes the form of an Expectation-Maximization (EM) algorithm and is based on a CRC approach that includes latent class modeling (cf. Baffour et al. (2021)). Furthermore, it is crucial to stratify job offers using occupation codes in accordance with the International Standard Classification of Occupations (ISCO). For example, in Poland, only the public register-based job portal ePraca reports these codes. To assign occupation codes to offers from other portals, one may use the classifier proposed by Beręsewicz et al. (2026). Given that any machine learning classifier introduces some degree of error, the direct inclusion of its predictions into subsequent modeling may lead to biased estimates. Therefore, the proposed method additionally allows for the correction of bias caused by classification errors in categorical variables. Finally, because job offers may not include the exact number of advertised vacancies, the developed approach accounts for missing and censored data.

The proposed algorithm allows researchers to estimate the number of vacancies both in general (e.g., for a country), as well as separately for all levels of the categorical covariates included in the model (e.g., for occupation codes). The approach is tested in a simulation study that utilizes

actual data from the Polish public register-based job portal ePraca. The simulation results show that the proposed algorithm outperforms a naïve two-step approach by substantially reducing estimation bias.

The paper is organized as follows. In Section 2, the proposed method is described. In Section 3, the results of a simulation study are presented. In Section 4, concluding remarks are provided. Technical details are presented in the Appendix.

2 Methodology

2.1 Notation and Data

We consider a finite population of job offers $U = \{1, 2, \dots, N\}$, where its size N is unknown, and data from three job portals. For $i \in U$, we define the following variables:

- $\mathbf{Y}_i = (Y_{i1}, Y_{i2}, Y_{i3}) \in \{0, 1\}^3$ – the capture history (Y_{i1} corresponds to a register-based portal that includes exact information about the occupation codes and the number of advertised vacancies, while the other two data sources can be commercial job portals),
- $G_i \in \{1, 2, \dots, C\}$ – the true (often unknown) occupation code,
- $X_i \in \{1, 2, \dots, K\}$ – the latent class (representing membership in a hidden population stratum),
- $V_i \in \{1, 2, \dots\}$ – the number of vacancies in a job offer.

We want to estimate the total number of vacancies, i.e.,

$$T = \sum_{i=1}^N V_i.$$

Units with $\mathbf{Y}_i = (0, 0, 0)$ are unobserved. Let S denote the set of observed job offers, i.e.,

$$S = \{i \in U : \mathbf{Y}_i \neq (0, 0, 0)\}.$$

Let

$$N_{\mathbf{y}, g, x} = \#\{i \in U : \mathbf{Y}_i = \mathbf{y}, G_i = g, X_i = x\}.$$

Furthermore, let $\tilde{G}_i$ denote the observed occupation code. If $Y_{i1} = 1$, then $\tilde{G}_i = G_i$. Otherwise, the observed code may differ from the true code.

Let $\tilde{V}_i$ denote the observed number of vacancies in offer i . If $Y_{i1} = 1$, then $\tilde{V}_i = V_i$. Otherwise, information about V_i might be missing or censored at 2+ (which corresponds to OJAs containing vague phrases such as “we will hire many people”).

Additionally, we assume the existence of an independent validation sample, denoted by S_{val} , which is disjoint from the main set of job offers S . For the units in S_{val} , both the true occupation code G_i and the observed (predicted) code $\tilde{G}_i$ are known. The role of S_{val} is to estimate the structure of misclassification errors.

The proposed approach is detailed in Sections 2.2 – 2.6 and illustrated in Figure 2.

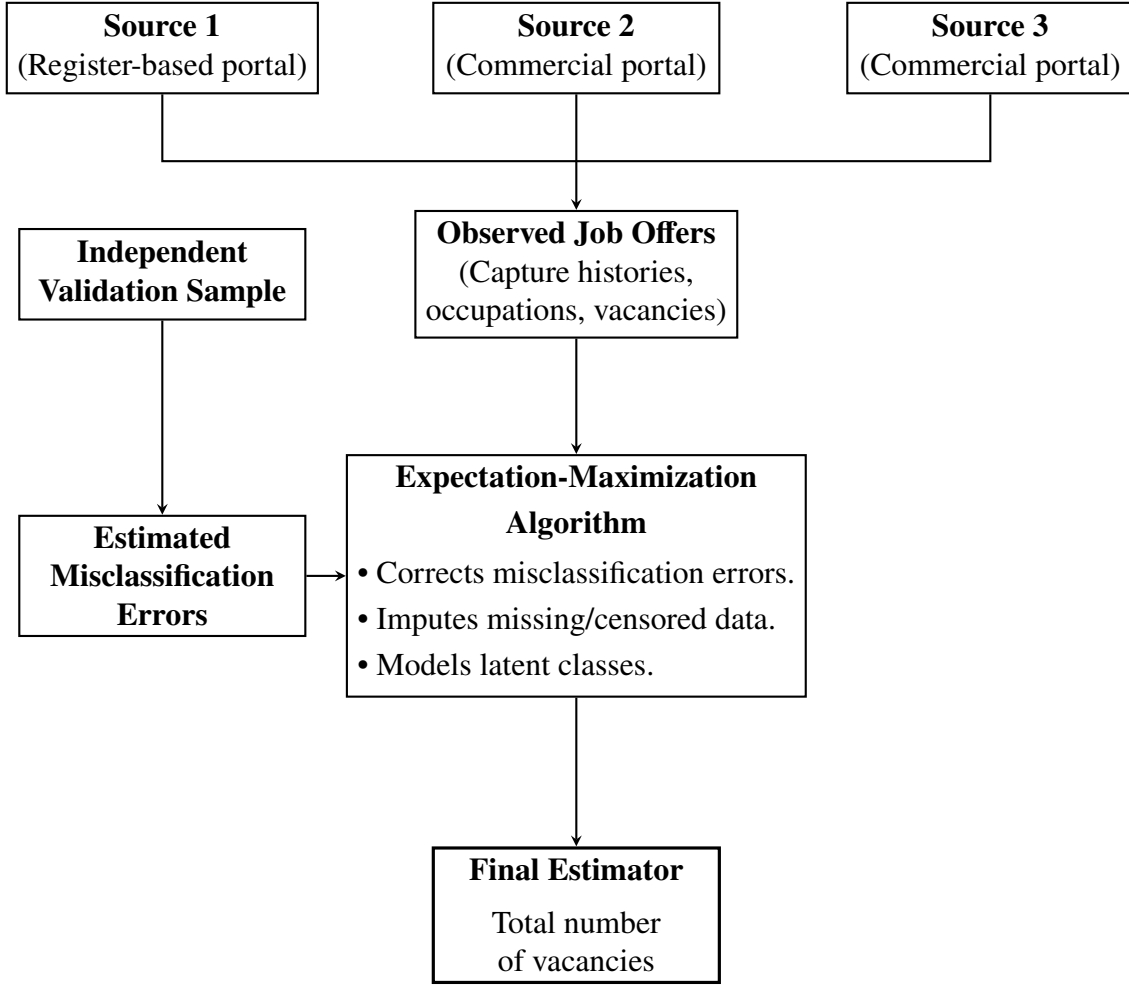


Figure 2: Flowchart of the proposed estimation method.

2.2 Assumptions

We assume the following:

- A. The population is closed.
- B. There are no linkage errors (future work will aim to relax this assumption).
- C. There exists a constant confusion matrix $\mathbf{\Pi}^{C \times C}$ such that for the units $i \in S$, we have

$$\mathbb{P}(\tilde{G}_i = h \mid G_i = g, Y_{i1} = 0) = \Pi_{gh}, \quad g, h \in \{1, 2, \dots, C\}.$$

- D. For each combination $(\mathbf{y}, g, x)$, there exists an expected cell frequency $m_{\mathbf{y}, g, x} > 0$, and the cell counts are independent and follow a Poisson distribution, i.e.,

$$N_{\mathbf{y}, g, x} \sim \text{Poisson}(m_{\mathbf{y}, g, x}).$$

We consider the following log-linear model (cf. Baffour et al. (2021)):

$$\log m_{\mathbf{y}, g, x} = \lambda + \lambda_g^G + \lambda_x^X + \sum_{s=1}^3 \lambda_{y_s}^{(s)} + \sum_{s=1}^3 \lambda_{y_s, x}^{(sX)} + \lambda_{g, x}^{(GX)} + \lambda_{y_1, y_2}^{(12)} + \lambda_{y_1, y_3}^{(13)}.$$

λ is the intercept and λ_g^G, λ_x^X are the main effects of the occupation code g and the latent class x , respectively. $\lambda_{y_1}^{(1)}, \lambda_{y_2}^{(2)}, \lambda_{y_3}^{(3)}$ are the main effects of the first, second, and third captures, respectively. The remaining terms represent the interactions between the corresponding main effects. Notably, we assume interactions between the register-based job portal and the commercial job portals, since listings on these platforms are frequently negatively correlated (i.e., an offer posted on a commercial portal is less likely to appear on the government one and vice versa).

- E. Given $G_i = g$, V_i follows a zero-truncated negative binomial distribution with parameters μ_g and σ (cf. Brännäs (1992)).
- F. Given $G_i = g$, the observation/coarsening mechanism generating $\tilde{V}_i$ is ignorable (missing/coarsened at random).
- G. Given $G_i = g$, V_i is independent of $\mathbf{Y}_i$ and X_i .
- H. Given $G_i = g$ and $\mathbf{Y}_i$, $\tilde{G}_i$ and $\tilde{V}_i$ are independent of each other and of X_i .

2.3 Estimation of the Confusion Matrix

Using the validation sample S_{val} , we calculate $\hat{\Pi}$ as an estimate of Π . Its elements are given by:

$$\hat{\Pi}_{g,\tilde{g}} = \frac{\#\{i \in S_{val} : G_i = g, \tilde{G}_i = \tilde{g}\}}{\#\{i \in S_{val} : G_i = g\}}.$$

Since $\hat{\Pi}$ is estimated externally from an independent validation sample, the subsequent EM algorithm is run conditional on $\hat{\Pi}$.

2.4 Likelihood Function

Let $f_g(\cdot; \mu_g, \sigma)$ denote the probability mass function of a zero-truncated negative binomial distribution with parameters μ_g and σ . For $i \in S$, we define $L_V(i \mid g; \mu_g, \sigma)$ as

- $f_g(\tilde{V}_i; \mu_g, \sigma)$, if $\tilde{V}_i = V_i$,
- $\mathbb{P}_{\mu_g, \sigma}(V_i \geq 2 \mid G_i = g)$, if $\tilde{V}_i = 2+$,
- 1, if $\tilde{V}_i$ is missing.

Let θ and ϕ denote the parameters associated with the log-linear model and with modeling the number of vacancies, respectively.

In this formulation, the capture part is treated through a complete-data log-likelihood based on the latent cell counts $N_{y,g,x}$, whereas the vacancy part is written directly in terms of the observed/coarsened vacancy data through $L_V(i \mid g; \phi)$. Therefore, conditional on $\hat{\Pi}$, the objective function used in the proposed EM algorithm consists of two components, given by:

$$l_c^{(Y)}(\theta) = \sum_{y,g,x} [N_{y,g,x} \log m_{y,g,x}(\theta) - m_{y,g,x}(\theta)] + \text{const}$$

and

$$l^{(V)}(\phi) = \sum_{i \in S} \log L_V(i \mid G_i; \phi).$$

2.5 EM Algorithm

In the E-step of the t -th iteration, we calculate

$$\tau_{i,g,x}^{(t)} = \mathbb{P}(G_i = g, X_i = x \mid \mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}})$$

and

$$N_{\mathbf{y},g,x}^{(t)} = \mathbb{E}[N_{\mathbf{y},g,x} \mid \{(\mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}].$$

Furthermore, let us define

$$Q^{(Y)}(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}) = \mathbb{E}[l_c^{(Y)}(\boldsymbol{\theta}) \mid \{(\mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}]$$

and

$$Q^{(V)}(\boldsymbol{\phi} \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}) = \mathbb{E}[l^{(V)}(\boldsymbol{\phi}) \mid \{(\mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}; \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}].$$

In the M-step, we compute

$$(\boldsymbol{\theta}^{(t+1)}, \boldsymbol{\phi}^{(t+1)}) = \left(\arg \max_{\boldsymbol{\theta}} Q^{(Y)}(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}), \arg \max_{\boldsymbol{\phi}} Q^{(V)}(\boldsymbol{\phi} \mid \boldsymbol{\theta}^{(t)}, \boldsymbol{\phi}^{(t)}, \hat{\boldsymbol{\Pi}}) \right).$$

The exact formulas and calculation methods for the E-step and M-step are presented in the Appendix.

The proposed algorithm can be summarized as follows.

1. **Input:** $\{(\mathbf{Y}_i, \tilde{G}_i, \tilde{V}_i)\}_{i \in S}, \hat{\boldsymbol{\Pi}}$.
2. **Initialization:** Initialize $N_{(0,0,0),g,x}^{(0)}$ and the weights $\tau_{i,g,x}^{(0)}$, and calculate the initial expected cell frequencies $N_{\mathbf{y},g,x}^{(0)}$ for $\mathbf{y} \neq (0,0,0)$.
3. For $t = 0, 1, \dots$, repeat until convergence:
 - a. **M-step:** Obtain $\boldsymbol{\theta}^{(t+1)}$ and $\boldsymbol{\phi}^{(t+1)}$ by solving the maximization tasks.
 - b. **E-step:** Calculate the new weights $\tau_{i,g,x}^{(t+1)}$ and update the expected cell frequencies $N_{\mathbf{y},g,x}^{(t+1)}$.
4. **Output:** The final estimated parameters $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$, along with the weights $\hat{\tau}_{i,g,x}$ and the expected cell frequencies $\hat{N}_{\mathbf{y},g,x}$.

2.6 Total Number of Vacancies Estimator

For $i \in S$, we define $V'_{i,g}$ as

- $\tilde{V}_i$, if $\tilde{V}_i = V_i$,
- $\mathbb{E}_{\hat{\boldsymbol{\phi}}}[V \mid V \geq 2, G = g]$, if $\tilde{V}_i = 2+$,
- $\mathbb{E}_{\hat{\boldsymbol{\phi}}}[V \mid G = g]$, if $\tilde{V}_i$ is missing.

We estimate the total number of vacancies as follows:

$$\hat{T} = \sum_{i \in S} \sum_{g=1}^C \sum_{x=1}^K \hat{\tau}_{i,g,x} V'_{i,g} + \sum_{g=1}^C \sum_{x=1}^K \hat{N}_{(0,0,0),g,x} \mathbb{E}_{\hat{\boldsymbol{\phi}}}[V \mid G = g].$$

3 Results

The proposed approach has been implemented in R (R Core Team, 2026), ensuring numerically stable optimization. To explore the method's utility, we conduct a simulation study using all

job offers available on ePraca in January 2026 as the target population. The considered variables are *code* and *vac*, which represent the first digit of the true occupation code and the number of vacancies in each job offer, respectively. One observation with the *code* value 0 is excluded from the analysis, since it refers to a military job offer. The resulting dataset has 10,524 records, and the total number of reported vacancies is 21,532.

First, we set a 9×9 confusion matrix $\mathbf{\Pi}$ to be used in the simulation study. Given that $G_i \in \{1, 2, \dots, 9\}$, the rows and columns of the matrix naturally represent the classification errors between the *code* values. To reflect real-world misclassification patterns, we adopt the confusion matrix reported by Beręsewicz et al. (2026, Table 11, Large model) as $\mathbf{\Pi}$.

The data generation mechanism is illustrated in Figure 3 and described in detail below:

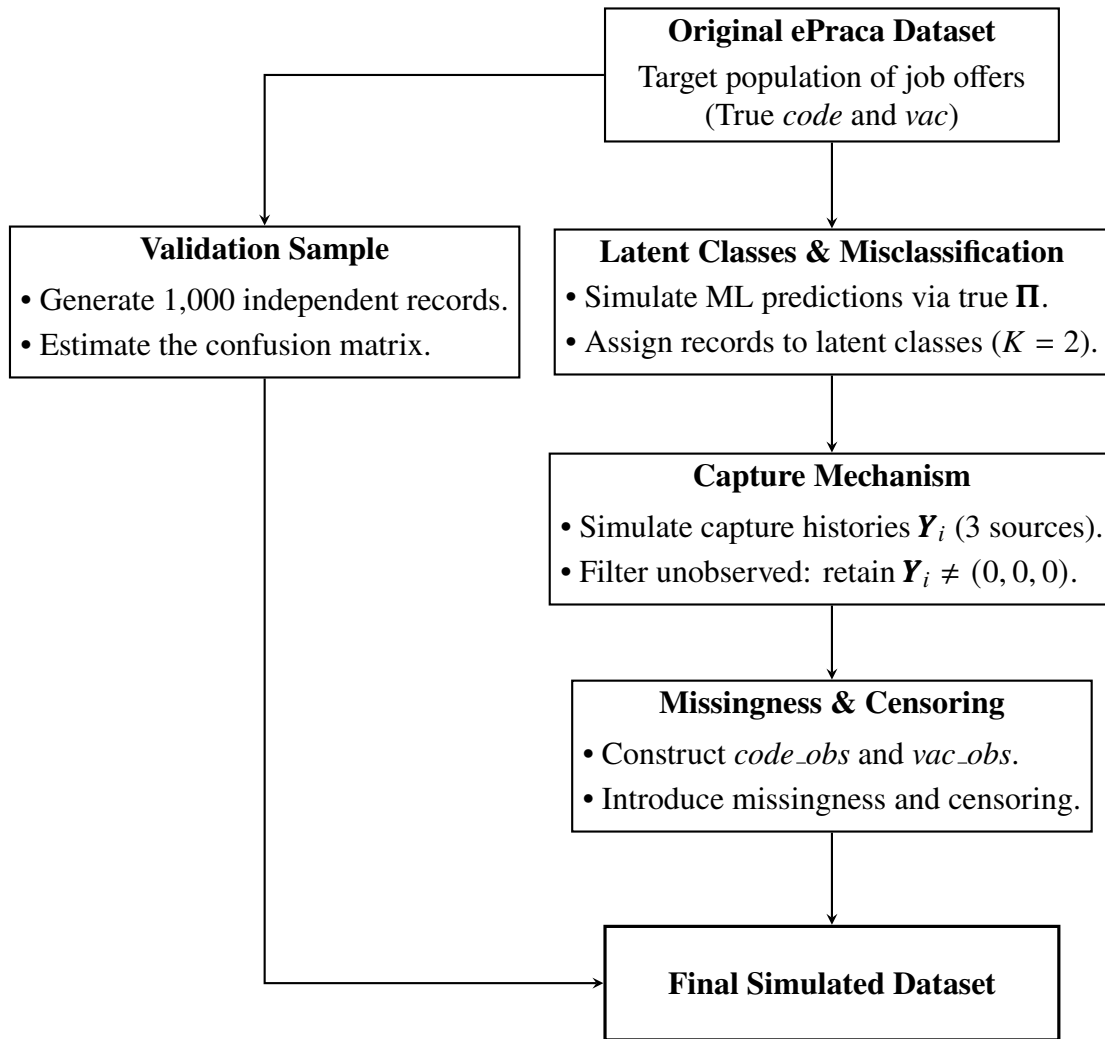


Figure 3: Flowchart of the data generation mechanism used in the simulation study.

1. For each record, we simulate the predicted occupation code, *code_{ml}*, based on $\mathbf{\Pi}$.
2. We calculate the empirical distribution of *code* in the population. Using this distribution, we generate a validation sample of size 1,000 with *code* values. For these newly generated records, we use $\mathbf{\Pi}$ to simulate the predicted occupation code. Then, we estimate $\mathbf{\Pi}$ based on the validation sample.

3. We consider two latent classes, i.e., $K = 2$. The probabilities of being in the second class (for each value of *code*) are defined by the vector

$$(0.4, 0.3, 0.2, 0.3, 0.4, 0.3, 0.2, 0.3, 0.4).$$

Then, following a Bernoulli distribution, the records are assigned to the latent classes based on these probabilities.

4. For the first class, the base capture probabilities are 0.8, 0.2, and 0.1, for the first, second, and third captures, respectively. For the second class, they are equal to 0.1, 0.8, and 0.6.

First, following a Bernoulli distribution, we simulate the first capture occasion. Then, if a record is captured, we decrease its base second and third capture probabilities by 0.05. Finally, we simulate the second and third capture occasions using the updated capture probabilities.

5. For the observed records, i.e., with $\mathbf{Y}_i \neq (0,0,0)$, we construct two new variables: *code_obs* (equivalent to $\tilde{G}_i$) and *vac_obs* (equivalent to $\tilde{V}_i$).

For the records captured in the first capture occasion, i.e., with $Y_{i1} = 1$, *code_obs* is equal to *code* and *vac_obs* is equal to *vac*. For the records not captured in the first source, we set *code_obs* equal to *code_ml*, and in terms of *vac_obs*, we proceed as follows:

- a. For each record, we sample *rand_val* from the uniform distribution $U(0,1)$. If $\text{rand_val} \leq 0.25$, we set *vac_obs* as missing.
- b. For each record with the true number of vacancies $\text{vac} \geq 2$ and $0.25 < \text{rand_val} \leq 0.5$, we set *vac_obs* as 2+.
- c. For the remaining records, we set *vac_obs* equal to *vac*.

We compare three estimation methods which are described below:

- A naïve two-step approach (*M1*): This approach does not include any bias correction and treats *code_obs* as ground truth. The number of job offers within a given occupation code is estimated using the method proposed by Baffour et al. (2021). Then, a zero-truncated negative binomial regression model that also accommodates censored data (cf. Brännäs (1992)) is applied to the observed data, i.e., with $\mathbf{Y}_i \neq (0,0,0)$. It is used to impute the missing and censored data. Subsequently, the average number of vacancies per offer within a given occupation code is estimated. These estimates are multiplied by the estimated strata sizes and summed, resulting in the final estimate of the total number of vacancies.
- The approach described in Section 2, but with the use of the true confusion matrix $\mathbf{\Pi}$ instead of its estimator (*M2*).
- The exact approach described in Section 2 (*M3*).

We conduct 100 simulations. To evaluate the results, we calculate the average estimate of the number of vacancies ($\hat{T}$), its mean bias (MB), mean relative bias (MRB), and root mean square error (RMSE). The results are presented in Table 1.

The simulation results demonstrate that the proposed method substantially decreases the bias of the naïve approach. Relatively small discrepancies between the bias of the models based on the estimated (*M3*) and true (*M2*) confusion matrices confirm the effectiveness of using an external validation sample. Although the proposed method slightly increases the RMSE compared to the naïve one, it substantially reduces the estimation bias. This increase in RMSE for the developed approach reflects a standard bias-variance trade-off. By correcting for

misclassification and modeling latent classes, the algorithm estimates additional parameters, which naturally causes higher variance in the final estimates.

Table 1: Average estimate of the number of vacancies, its mean bias, relative mean bias, and root mean square error over 100 simulations.

Method	$\hat{T}$	MB	MRB	RMSE
<i>M1</i>	20,937.32	-594.68	-2.76%	1,234.53
<i>M2</i>	21,597.01	65.01	0.30%	1,286.64
<i>M3</i>	21,648.32	116.32	0.54%	1,370.56

Note. The true total number of vacancies in the target population is $T = 21,532$. $\hat{T}$ represents the average estimate of the number of vacancies over 100 simulations. *M1* represents the naïve two-step approach without bias correction; *M2* is the proposed EM algorithm utilizing the true confusion matrix $\mathbf{\Pi}$; *M3* is the proposed EM algorithm using the estimated confusion matrix $\hat{\mathbf{\Pi}}$. MB: Mean Bias; MRB: Mean Relative Bias; RMSE: Root Mean Square Error.

Additionally, Table 2 (included in the Appendix) presents the results of the simulation separately for all values of the *code* variable. In most cases, these results demonstrate the improved performance of the proposed method over the naïve one.

In the spirit of open science and to ensure the transparency and reproducibility of the findings, the simulation code and data are publicly available in the GitHub repository at <https://github.com/OJALAB/submission-2026-IAOS-prize>.

4 Conclusions

In this paper, a proof-of-concept method for estimating the number of vacancies has been developed. It takes the form of an EM algorithm and is based on a CRC approach that includes latent class modeling. It corrects for the bias associated with classification errors in the included categorical covariates and accounts for missing and censored data. The simulation results show that the proposed algorithm outperforms the naïve two-step method by substantially reducing estimation bias.

For simplicity of description, only one categorical covariate was considered. However, the framework can be easily extended in several ways:

1. Adding more data sources. Researchers can readily incorporate 4 or even 5 job portals.
2. Modifying the interaction terms between data sources (i.e., adding or excluding them). For example, there might be two commercial job portals that cover a similar part of the labor market, thereby leading to a positive correlation between the captures in these two data sources.
3. Relaxing the assumption that in one data source we observe the true values of occupation codes and the number of vacancies. There is no need for an error-free dataset.
4. Adding more categorical covariates (both with and without measurement error). Here, one can also consider incorporating administrative data, e.g., characteristics of entities or information about their locations.
5. Applying alternative count distributions for the number of vacancies, e.g., a zero-truncated Poisson distribution.
6. Using different censoring thresholds. If job offers with phrases such as “we will hire at least k people” are observed, they can be directly included in the algorithm as $k+$.

There are also aspects that warrant further research. A key challenge concerns addressing linkage errors. Another area that merits further investigation is incorporating the sampling

uncertainty associated with the estimation of the confusion matrix. Finally, as presented by Lewandowski (2026), future labor market analysis should focus on skills rather than occupations. Therefore, subsequent adjustments of the proposed method could incorporate skills classification.

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Appendix

E-step

In the E-step, we calculate $\tau_{i,g,x}^{(t)}$ as follows:

$$\tau_{i,g,x}^{(t)} = \frac{\mathbb{P}(\tilde{G}_i | G_i = g, \mathbf{Y}_i; \hat{\boldsymbol{\Pi}}) \cdot L_V(i | g; \boldsymbol{\phi}^{(t)}) \cdot m_{\mathbf{Y}_i, g, x}(\boldsymbol{\theta}^{(t)})}{\sum_{g'=1}^C \sum_{x'=1}^K \mathbb{P}(\tilde{G}_i | G_i = g', \mathbf{Y}_i; \hat{\boldsymbol{\Pi}}) \cdot L_V(i | g'; \boldsymbol{\phi}^{(t)}) \cdot m_{\mathbf{Y}_i, g', x'}(\boldsymbol{\theta}^{(t)})}.$$

Note that $\mathbb{P}(\tilde{G}_i | G_i = g, \mathbf{Y}_i; \hat{\boldsymbol{\Pi}})$ equals $\hat{\Pi}_{g, \tilde{G}_i}$ if $Y_{i1} = 0$, and equals 1 if $Y_{i1} = 1$ and $\tilde{G}_i = g$ (and 0 otherwise).

For the units with $\mathbf{y} \neq (0,0,0)$, we have

$$N_{\mathbf{y}, g, x}^{(t)} = \sum_{i \in S: Y_i = \mathbf{y}} \tau_{i, g, x}^{(t)},$$

whereas if $\mathbf{y} = (0,0,0)$, we have

$$N_{\mathbf{y}, g, x}^{(t)} = m_{\mathbf{y}, g, x}(\boldsymbol{\theta}^{(t)}).$$

M-step

In the M-step, the first maximization task (i.e., finding $\boldsymbol{\theta}^{(t+1)}$) is equivalent to fitting a Poisson Generalized Linear Model with a logarithmic link function to the contingency table containing the values $\{N_{\mathbf{y}, g, x}^{(t)}\}$. For the second maximization task (i.e., calculating $\boldsymbol{\phi}^{(t+1)}$), we maximize the expected vacancy log-likelihood contribution based on exact, censored, and missing outcomes, i.e.,

$$\boldsymbol{\phi}^{(t+1)} = \arg \max_{\boldsymbol{\phi}} \sum_{i \in S} \sum_{g=1}^C \sum_{x=1}^K \tau_{i, g, x}^{(t)} \log L_V(i | g; \boldsymbol{\phi}).$$

Results for occupation codes

Table 2: Average estimates of the number of vacancies by occupation code, together with mean biases, relative mean biases, and root mean square errors over 100 simulations.

<i>code</i>	T_g	Method	$\hat{T}_g$	MB	MRB	RMSE
1	301	<i>M1</i>	344.51	43.51	14.46%	56.72
		<i>M2</i>	305.50	4.50	1.49%	27.80
		<i>M3</i>	306.34	5.34	1.77%	39.51
2	2,784	<i>M1</i>	2,798.77	14.77	0.53%	145.06
		<i>M2</i>	2,802.14	18.14	0.65%	172.50
		<i>M3</i>	2,808.89	24.89	0.89%	186.56
3	2,587	<i>M1</i>	2,528.49	-58.51	-2.26%	245.52
		<i>M2</i>	2,591.16	4.16	0.16%	286.62
		<i>M3</i>	2,598.25	11.25	0.43%	311.08
4	1,403	<i>M1</i>	1,320.07	-82.93	-5.91%	143.52
		<i>M2</i>	1,368.43	-34.57	-2.46%	135.94
		<i>M3</i>	1,380.39	-22.61	-1.61%	148.50
5	2,466	<i>M1</i>	2,341.01	-124.99	-5.07%	170.91
		<i>M2</i>	2,398.15	-67.85	-2.75%	160.12
		<i>M3</i>	2,410.21	-55.79	-2.26%	163.96
6	96	<i>M1</i>	103.61	7.61	7.93%	17.71
		<i>M2</i>	102.70	6.70	6.98%	30.49
		<i>M3</i>	111.85	15.85	16.51%	62.18
7	6,719	<i>M1</i>	6,694.72	-24.28	-0.36%	430.78
		<i>M2</i>	6,737.63	18.63	0.28%	484.67
		<i>M3</i>	6,752.68	33.68	0.50%	519.31
8	3,165	<i>M1</i>	2,926.63	-238.37	-7.53%	331.22
		<i>M2</i>	3,292.76	127.76	4.04%	336.33
		<i>M3</i>	3,283.16	118.16	3.73%	320.10
9	2,011	<i>M1</i>	1,879.50	-131.50	-6.54%	190.39
		<i>M2</i>	1,998.56	-12.44	-0.62%	167.16
		<i>M3</i>	1,996.55	-14.45	-0.72%	181.72

Note. T_g is the total number of vacancies for the *code* value g ; $\hat{T}_g$ represents the average estimate of T_g over 100 simulations. *M1* represents the naïve two-step approach without bias correction; *M2* is the proposed EM algorithm utilizing the true confusion matrix $\mathbf{\Pi}$; *M3* is the proposed EM algorithm using the estimated confusion matrix $\hat{\mathbf{\Pi}}$. MB: Mean Bias; MRB: Mean Relative Bias; RMSE: Root Mean Square Error.